

CN Capstone Project

Mac 1 - Private DNS Server and Client Documentation

Phase 1 / Detailed Evidence Record

Team: Packet Forge Domain: cn-capstone.test Machine: Mac 1 - cn-dns

1. Scope of This Mac 1 Document

Mac 1 provides the project private DNS service and is also used as a client. This report documents only the evidence associated with Mac 1: dnsmasq service/configuration, private DNS resolution, the public-resolver comparison, the final HTTPS client test, and the Wireshark captures supplied for Mac 1. Backend, nginx, and other-machine evidence is referenced only to explain the project path.

2. Correct Project Topology

Machine	Project role / hostname	IPv4	Interface	Mask / gateway	Service
Mac 1	cn-dns - Private DNS + client	10.7.9.245	en0	255.255.224.0 / 10.7.0.1	dnsmasq :53
Mac 2	cn-edge - HTTPS edge / reverse proxy / LB	10.7.5.53	en0	255.255.224.0 / 10.7.0.1	nginx HTTPS :8443
Mac 3	Backend A	10.7.22.10	en0	255.255.224.0 / 10.7.0.1	HTTP :3001
Mac 4	Backend B + client	10.7.7.25	en0	255.255.224.0 / 10.7.0.1	HTTP :3002

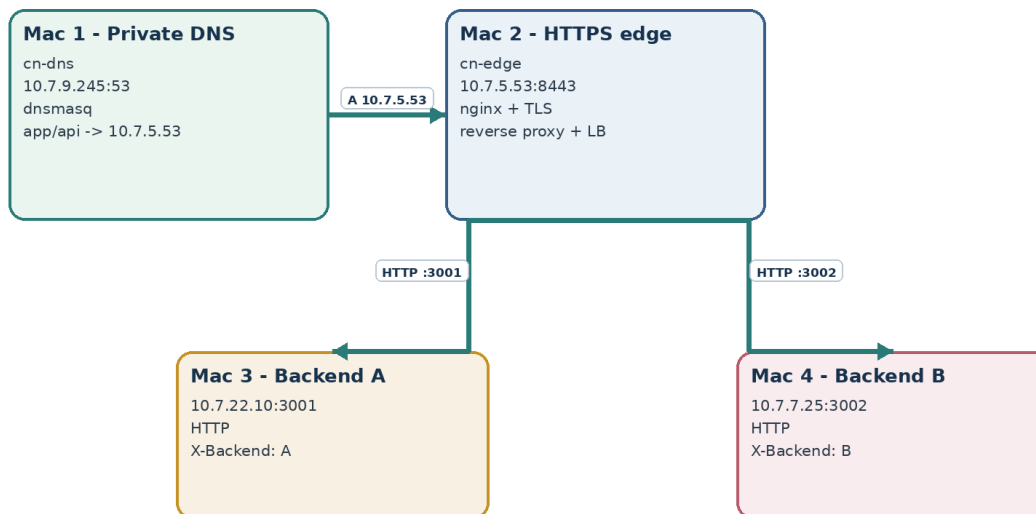
Canonical final addressing used throughout the project: Mac 1 10.7.9.245, Mac 2 10.7.5.53, Mac 3 10.7.22.10, Mac 4 10.7.7.25. The subnet mask is 255.255.224.0 (/19) and the gateway is 10.7.0.1.

3. Mac 1 Identity and Responsibilities

- Hostname: cn-dns.
- Active interface: en0.
- IPv4 address: 10.7.9.245.
- Subnet mask: 255.255.224.0 (/19).
- Default gateway: 10.7.0.1.
- Private DNS service: dnsmasq on UDP/TCP port 53.
- Project records: app.cn-capstone.test -> 10.7.5.53 and api.cn-capstone.test -> 10.7.5.53.

4. Project Topology Diagram

Phase 1 project service topology - Mac 1 view



Mac 1 provides the private .test DNS record; Mac 2 is the single HTTPS entry point.

Figure 1. Spaced service-topology diagram. The diagram separates the DNS service, HTTPS edge, and two backend branches.

5. dnsmasq Configuration

The supplied Mac 1 evidence shows dnsmasq bound to localhost and the LAN address, using en0, and serving the two project records.

```
adityarana@cn-dns ~ % echo "===== MAC 1 - DNS CONFIGURATION ====="
echo
echo "----- MAIN DNSMASQ CONFIG -----"
cat /opt/homebrew/etc/dnsmasq.conf

echo
echo "----- PROJECT DNS RECORDS -----"
cat /opt/homebrew/etc/dnsmasq.d/project.conf
===== MAC 1 - DNS CONFIGURATION =====

----- MAIN DNSMASQ CONFIG -----
port=53
listen-address=127.0.0.1,10.7.9.245
bind-interfaces
no-resolv
server=8.8.8.8
domain-needed
bogus-priv
cache-size=1000
local-ttl=30
conf-dir=/opt/homebrew/etc/dnsmasq.d,*.conf

----- PROJECT DNS RECORDS -----
address=/app.cn-capstone.test/10.7.5.53
address=/api.cn-capstone.test/10.7.5.53
adityarana@cn-dns ~ %
```

Figure 2. Supplied Mac 1 terminal evidence showing the dnsmasq configuration and project A records.

6. DNS Resolution

```
adityarana@cn-dns ~ % echo "===== MAC 1 - DNS RESOLUTION ====="
echo
echo "----- app.cn-capstone.test -----"
dig @127.0.0.1 +short app.cn-capstone.test

echo
echo "----- api.cn-capstone.test -----"
dig @127.0.0.1 +short api.cn-capstone.test

echo
echo "----- LAN DNS ADDRESS -----"
dig @10.7.9.245 +short app.cn-capstone.test
===== MAC 1 - DNS RESOLUTION =====

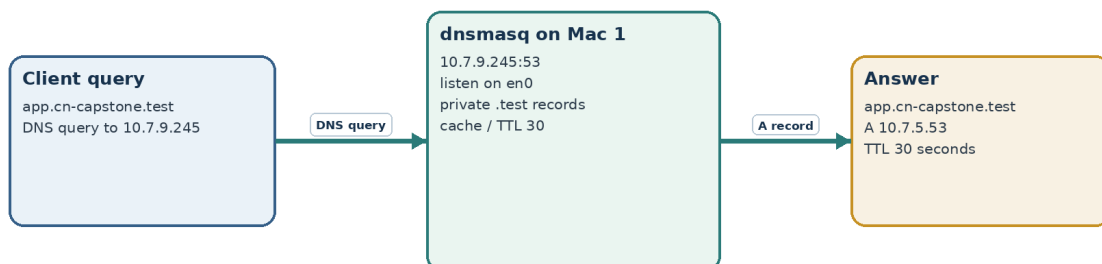
----- app.cn-capstone.test -----
10.7.5.53

----- api.cn-capstone.test -----
10.7.5.53

----- LAN DNS ADDRESS -----
10.7.5.53
adityarana@cn-dns ~ %
```

Figure 3. Supplied Mac 1 DNS resolution evidence. Both project hostnames resolve to the Mac 2 edge address 10.7.5.53.

Private DNS resolution path



The private .test name resolves only through the project DNS service.

Figure 4. Spaced private-DNS resolution diagram.

Public resolver comparison

The public resolver does not provide the private project record. The supplied evidence shows an NXDOMAIN response from 8.8.8.8 for app.cn-capstone.test.

```
● ● ● Public resolver comparison

$ dig @8.8.8.8 app.cn-capstone.test
status: NXDOMAIN
SERVER: 8.8.8.8#53(8.8.8.8)

Interpretation:
The private .test record is intentionally not public.
```

Figure 5. Public DNS comparison used to show that the .test record is private.

7. DNS Service and Port 53

```
adityarana@cn-dns ~ % echo "==== MAC 1 - DNS SERVICE ====="

echo
echo "----- DNSMASQ SERVICE -----"
brew services list | grep dnsmasq

echo
echo "----- DNS PORT 53 -----"
[sudo lsof -nP -iUDP:53 -iTCP:53
==== MAC 1 - DNS SERVICE =====

----- DNSMASQ SERVICE -----
dnsmasq    none          root

----- DNS PORT 53 -----
[Password:
COMMAND PID   USER   FD   TYPE             DEVICE SIZE/OFF NODE NAME
dnsmasq 868  nobody  4u   IPv4 0x9e0b5371cf46d8fb      0t0  UDP 10.7.9.245:53
dnsmasq 868  nobody  5u   IPv4 0x5dc4db2cd751f42b      0t0  TCP 10.7.9.245:53 (LISTEN)
dnsmasq 868  nobody  6u   IPv4 0xc5aec416178b1e9b      0t0  UDP 127.0.0.1:53
dnsmasq 868  nobody  7u   IPv4 0xbb1fdbbbc505db672     0t0  TCP 127.0.0.1:53 (LISTEN)
adityarana@cn-dns ~ %
```

Figure 6. Supplied Mac 1 service evidence. dnsmasq is shown listening on port 53 on the project address and localhost.

This proves that Mac 1 is not merely configured with DNS records; it is actively providing the DNS service used by the project.

8. Final HTTPS Client Test

```
10.7.9.30
adityarana@cn-dns ~ % echo "==== MAC 1 - FINAL HTTPS TEST ====="

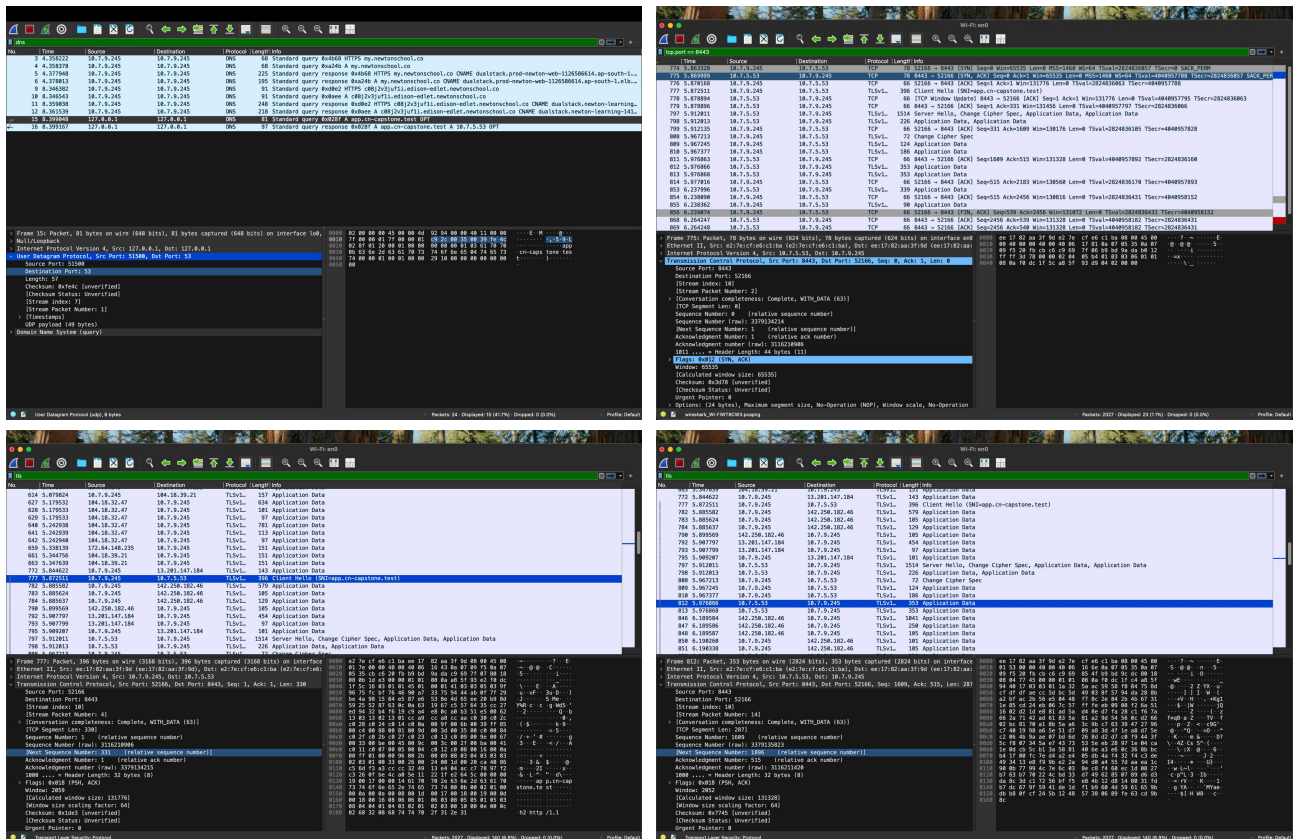
[curl -i https://app.cn-capstone.test:8443/api/status
==== MAC 1 - FINAL HTTPS TEST =====
HTTP/1.1 200 OK
Server: nginx/1.31.6
Date: Thu, 01 Oct 2026 10:05:22 GMT
Content-Type: application/json
Content-Length: 56
Connection: keep-alive
X-Backend: A
Cache-Control: max-age=60

{"backend": "A", "status": "healthy", "server": "Mac 3"}%
adityarana@cn-dns ~ %
```

Figure 7. Supplied Mac 1 final HTTPS test using the project hostname. The response is HTTP 200, with X-Backend: A and Cache-Control: max-age=60.

9. Wireshark Evidence Captured on Mac 1

Task G was not performed as part of the Mac 2 role. The supplied Mac 1 evidence contains the packet captures used for the Mac 1 Wireshark section, so the packet-level evidence is documented here rather than duplicated in the Mac 2 report.



Figures 8-11. Supplied Mac 1 Wireshark captures covering DNS, TCP/TLS establishment, TLS handshake, and encrypted application data.

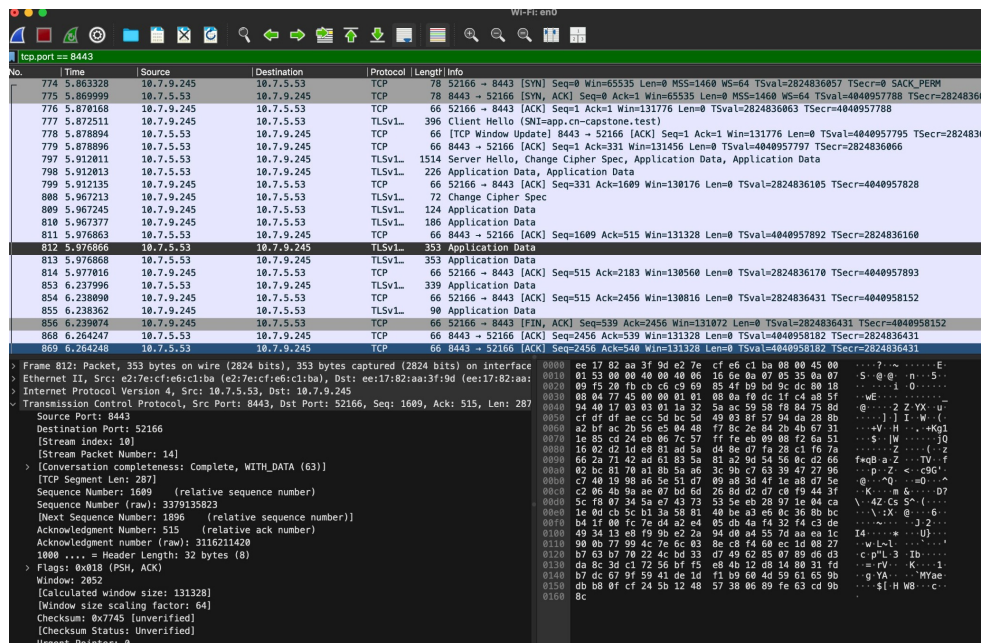


Figure 12. Supplied Mac 1 packet-flow evidence showing the established TCP/TLS exchange and encrypted application data.

10. What Mac 1 Proves

Evidence	Mac 1 result	Status
Machine identity	cn-dns / 10.7.9.245 / en0	Verified
dnsmasq service	Port 53 active	Verified
Project records	app/api -> 10.7.5.53	Verified
Private DNS resolution	Local project names resolve through 10.7.9.245	Verified
Public resolver comparison	8.8.8.8 returns NXDOMAIN for app.cn-capstone.test	Verified
Final HTTPS client test	HTTPS to app.cn-capstone.test:8443 returns HTTP 200	Verified
Wireshark	DNS, TCP, TLS and encrypted-data captures in supplied Mac 1 evidence	Documented

11. Mac 1 Final Status

- Private DNS is configured and serving the project .test records.
- The project hostnames resolve to the Mac 2 edge address 10.7.5.53.
- The final client HTTPS test reaches the application through the project hostname.
- The supplied Mac 1 packet captures document the DNS, TCP, TLS, and encrypted-data stages.
- This report intentionally does not claim nginx configuration, backend configuration, or Mac 2 Wireshark work was performed on Mac 1.